

CHENNAI INSTITUTE OF TECHNOLOGY

(Autonomous)

CS5305 - MACHINE LEARNING

TITLE OF THE PROJECT

ParkPilot - AI-Powered Intelligent Parking Space Detection and Navigation Using Computer Vision

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Parking in urban areas is a persistent challenge, with drivers wasting significant time and fuel searching for available spaces. ParkPilot is a Computer Vision and Machine Learning based system that detects vacant and occupied parking spaces in real time from camera feeds and guides drivers to the nearest available slot. The system uses object detection models to identify vehicles and classify parking slot occupancy, improving parking efficiency and reducing congestion.

I. Introduction (1/1)

Urban parking management is a growing concern as vehicle ownership increases faster than the availability of parking infrastructure. Drivers often circle around crowded areas searching for a free spot, which leads to wasted time, increased fuel consumption, and traffic congestion.

The proposed project, ParkPilot, is a Machine Learning and Computer Vision based system that detects available and occupied parking spaces in real time from live camera feeds. The system uses object detection models to identify vehicles within marked parking slots and classifies each slot as vacant or occupied, then guides the driver to the nearest free slot through a navigation interface.

This project aims to reduce the time spent searching for parking, ease traffic congestion around parking facilities, and enable smarter, more efficient use of existing parking infrastructure.

II. Problem Statement (1/1)

Finding an available parking space in crowded urban areas, malls, and office complexes remains a significant challenge. Existing sensor-based parking solutions are often expensive to install and maintain, requiring dedicated hardware for every individual parking slot.

There is a need for a cost-effective, camera-based solution that can accurately detect free and occupied parking spaces in real time and guide drivers to the nearest available slot without the need for per-slot hardware sensors. Such a system can reduce search time, lower fuel wastage and emissions, and ease traffic congestion caused by vehicles circling in search of parking.

III. Project Description (1/1)

ParkPilot is a real-time parking space detection and navigation system that identifies vacant and occupied parking slots from camera feeds using Computer Vision and Machine Learning techniques.

Technologies Used:

Programming Language: Python

Computer Vision Framework: OpenCV

Object Detection Model: YOLOv8

Deep Learning Framework: PyTorch / TensorFlow

Navigation & Mapping: Custom slot-grid mapping module

Working Process:

1. Capture live video feed from parking lot cameras.
2. Detect parking slot boundaries and vehicles using YOLOv8.
3. Classify each slot as vacant or occupied.
4. Track slot status changes in real time.
5. Identify the nearest available slot to the entry point.
6. Guide the driver via a navigation display / app interface.
7. Update occupancy status continuously on a live dashboard.

Designed to operate efficiently using standard IP/CCTV camera hardware.

IV. Aim & Scope of the Proposed Work (1/1)

Aim:

To develop a real-time, camera-based parking space detection and navigation system that accurately identifies available parking slots and guides drivers to them, thereby reducing search time and easing traffic congestion.

Scope:

- Real-time detection of vacant and occupied parking spaces.
- Navigation guidance to the nearest available slot.
- Implementation using lightweight, camera-based Computer Vision techniques.
- Support for multi-camera and multi-level parking layouts.
- Extension to mobile and web-based navigation applications.
- Integration with malls, offices, airports, and smart city infrastructure.
- Reduction of traffic congestion and fuel wastage from parking search.

Expected Outcome:

A user-friendly system capable of detecting real-time parking availability and guiding drivers to the nearest free slot, improving parking efficiency and reducing urban traffic congestion.